Smaller existential definitions by shared elliptic witnesses

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Abstract

Sun's simultaneous prime-pattern theorem allows the local elliptic witnesses in our integer definition to be shared. We obtain an explicit, compiled polynomial with substantially fewer variables and a shorter circuit record. A separate construction gives a conditional forty-variable definition of the actual subset $\mathbb{Z} \subset \mathbb{Q}$. The latter uses a new scalar certificate and finite-algebra elimination; its coefficients have not been generated. All integer-definition assertions retain the arithmetic hypotheses stated below. The accompanying audits check the new deductions, the relevant analytic arguments, and the recorded circuit; they are not formal verification of the underlying preprints.

1 The two conclusions

The shared-witness method gives two different improvements. The first is a fully recorded polynomial, using the constants already present in the earlier construction. The second is a forty-variable conditional theorem, whose remaining fixed constants and finite-algebra elimination have not been numerically compiled. These are definitions of the actual integers:

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{Y} \in \mathbb{Q}^r \quad P(X, \mathbf{Y}) = 0.$$

The free input X is not included in the witness count r.

The accompanying package records the measured circuits and their independent recounts. Degree means a proved upper bound obtained by propagation through the polynomial circuit. Circuit bytes count a lossless encoding with the same format and decoder for every variant. The fixed coefficient $3^{2^{24}}$ is recorded by its twenty-four-squaring recipe in every case. These are circuit-description lengths, not expanded monomial-list lengths. The forty-variable theorem supplies no competitive degree or bit-length bound by itself.

Construction	Witnesses	Degree bound	Circuit bytes
Earlier recorded polynomial	1,066	227,772	802,275
Shorter new circuit	100	596,676	33,971
Smaller new witness count	82	7,190,904	47,554
Finite-algebra construction	40	not computed	not emitted

The 100-variable circuit is about 23.6 times shorter in this encoding. Under the same `xz` compression, its record occupies $18,412$ bytes, against $327,632$ bytes before. A 104-variable variant compresses to $18,368$ bytes, a difference of only 44 bytes; we retain the 100-variable version as the simpler variable count. The 82-variable version has larger degree and circuit size. The source package provides both principal variants, their builders, decoders and independent counts. The complete human-readable recipes occupy $5,410$ ASCII characters for the 100-variable version and $5,337$ for the 82-variable version, including whitespace. The latter contains 269 digit characters. This textual saving also reflects shorter exposition; the circuit comparison above uses a fixed encoding and measures the mathematical simplification more directly.

2 An explicit definition with eighty-two witnesses

This construction retains the arithmetic inputs and explicit constants of [2], and additionally assumes [1, Theorem 4.1]. It uses the full preparation predicate from the earlier construction; no estimate with preparation parameter zero is needed. Put

$$F = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2, \quad E : y^2 = x(x^2 - D),$$

$$P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right), \qquad 2E(F) = \mathbb{Z}P.$$

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The inherited bounds are

$$h(x(nP)) \geq n^2 - 6, \qquad h(a) \leq HM(a)^{24}, \qquad \log_2 H < 4783227, \tag{1}$$

where M is the odd-contact radical outside the primes at most 76176 for $\infty, 0, 1, 2401, -343/32$. All uses of these bounds below have the same exceptional set.

2.1 The polynomial blocks

Represent a nonzero point of $2E(F)$ by $u, v \in F$ with

$$v^2(4 - v^2) = 4Du^4, \qquad u \neq 0. \tag{2}$$

This chart uses four rational variables and has exactly the stated image. Write $t = (v^2 - 2)/2$, so that

$$x = u^{-2}, \qquad f = 1/x = u^2, \qquad z = -x/y = -u/t.$$

Use the earlier fixed rational expressions for $f(X - [4]Q)$ and $z(X - [4]Q)$, including their values zero at O, and the polynomial

$$\ell(Z) = Z \sum_{j=0}^{16} \frac{(-1)^j \binom{2j}{j}}{4j + 1} (DZ^4)^j.$$

These are compositions of fixed polynomials; they introduce no variables. The chart identities and nonvanishing assertions are those of [2]. In particular $\ell(z(Q)) \neq 0$ for $Q \neq O$.

For rational b, W, define $\mathrm{Par}(b, W)$ using variables a, e, w and two square conditions. Put

$$U = \frac{b(1 - ba^2)}{1 + ba^2}, \quad B = 17b,$$

$$A_1 = B(B - e)(B + 3e),$$
$$A_2 = B\{w^2 - e[B - (WU)^2]\}\{w^2 + 3e[B - (WU)^2]\}.$$

Require $1 + ba^2 \neq 0$ and $A_1, A_2 \in \mathbb{Q}^2$. The two square conditions use just one additional variable q, through

$$(A_1^2 + A_2^2)\{1 - 2(A_1 + A_2)q^2 + (A_1 - A_2)^2 q^4\} = 0. \tag{3}$$

Indeed, unless $A_1 = A_2 = 0$, put $r = 1/q$; the square roots are $(r^2 + A_1 - A_2)/(2r)$ and $(r^2 - A_1 + A_2)/(2r)$. Conversely choose signs of the square roots with nonzero sum and take its reciprocal. Thus Par uses four rational witnesses.

We use the two established properties of this block: $\mathrm{Par}(b, W)$ excludes negative odd valuations of bW^2 outside $2, 3, 17$ when $W \neq 0$; it holds whenever bW^2 is positive, has square reduced denominator, and is a square over $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. These are the two properties required below; the accompanying algebra audit also verifies them directly for the displayed block.

For field tests (D_j, k_j) and $r \neq 0$ put

$$W_F = \prod_j \left(1 + \frac{D_j^2}{r^{2k_j - 1}}\right), \qquad \Theta = \mathrm{Tr}_{F/\mathbb{Q}}(W_F^{-1}), \qquad \mathrm{Par}(r, \Theta^{-1}), \tag{4}$$

retaining the nonzero guards for W_F and Θ. At an odd valuation $\alpha = v_p(r)$, each factor has valuation $\min(0, 2v(D_j) - (2k_j - 1)\alpha)$. If λ is the maximum of $v(W_F)$ over the places above p, then $v_p(\Theta) \geq -\lambda \geq 0$. Parity soundness forces $\alpha > 0$ and $\lambda \geq -\alpha/2$; at a place attaining λ every test satisfies $v(D_j) \geq k_j$. This is also valid at inert unramified primes. Conversely, if $v_p(r) = 1$ and the tests hold at one place over a split prime $p > 2$, the two inverse products have residues 0 or 1, with at least one 1. Hence Θ is a p-adic unit. For $r > 0$ all factors are positive at both real embeddings, so W_F and Θ are nonzero. These are precisely the single-group assertions of [1, Lemma 3.1], with the full preparation predicate substituted.

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2.2 The formula

Put $\beta = X^2 + 2$ and quantify chart points

$$A, S, T, Z, J_0, X_0, X_1, X_2, X_3, X_4.$$

Set $(Q_0, Q_1, Q_2, Q_3, Q_4) = (P, A, S, T, Z)$, $Y_i = X_i - [4]Q_i$, and require $Y_0 \neq O$. For $1 \leq i \leq 4$ put

$$L_i = \frac{\ell(z(X_i))}{\ell(z(X_0))}, \qquad L_i' = \frac{\ell^\sigma(z(Y_i)^\sigma)}{\ell^\sigma(z(Y_0)^\sigma)}.$$

Let

$$\mathcal{F}(a, b, c) = b^2 - a(a-2)bc + (a-2)^2c^2 - 1,$$

and use these sixteen field tests:

$$\begin{array}{lll} f(X_i),\ f(Y_i)^\sigma & (0 \leq i \leq 4), & k = 1, \\ L_i - L_i' & (1 \leq i \leq 4), & k = 64, \\ L_1 - \beta, \quad \mathcal{F}(L_2, L_3, L_4) & & k = 64. \end{array} \tag{5}$$

Let $a_1, \ldots, a_8$ be the rational coefficients of $x(A), x(S), x(T), x(Z)$. Quantify one rational η, and set

$$t_0 = \eta, \qquad t_i = a_i - \eta \quad (1 \leq i \leq 8),$$

$$R(t) = t(t-1)(t-2401)(32t+343)(2t-1), \qquad r_0 = R(t_0), \quad r_i = -R(t_i)\ (i \geq 1).$$

Require all $r_i \neq 0$ and apply (4) to (5) for each r_i.

Retain the earlier scalar-height block and anchor, as follows. Choose μ among the 256 signed squarefree divisors of $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, using its earlier one-variable polynomial selector. With $r = R(\beta)$, form the rational product

$$W_0 = \prod_{(B,k)} (1 + B^2/r^{2k-1}),$$

where the six B are the two rational coefficients of each of $f(J_0), f(S), z(S)/z(J_0)$, with orders $1, 1, 48$, respectively. Require $\mathrm{Par}(\mu r, W_0)$. Finally, with $C = 3^{2^{24}}$, impose

$$\mathrm{Nm}(u_S)^2 + C^2(c_1^2 + 3c_2^2 + 3c_3^2 - 4) = 0. \tag{6}$$

The constant C is recorded by twenty-four squarings, not by quantified variables. All rational-expression denominators are guarded as in the supplied builder.

2.3 Local comparison and soundness

Write $Q_i = n_i P$, $X_0 = hP$, and $X_i = m_i P$. At a place supplied by (4), the formal-group comparison and the relative precision of ℓ give

$$L_i = \frac{m_i}{h}(1 + \epsilon_i), \qquad L_i' = \frac{m_i - 4n_i}{h - 4}(1 + \epsilon_i'), \qquad v(\epsilon_i), v(\epsilon_i') \geq 64.$$

Zero numerators are interpreted directly. The logarithm denominators ensure $h \neq 0, 4$. At least one of $h, h-4$ is a p-adic unit; its exact ratio, hence its truncated ratio, is integral. The test $L_i - L_i'$ makes the other truncated ratio integral as well. Both exact ratios are therefore integral, and both approximation errors have absolute valuation at least 64. Consequently

$$\frac{4(hn_i - m_i)}{h(h-4)} \in p^{64}\mathbb{Z}_p, \qquad L_i \equiv n_i \pmod{p^{64}}.$$

It follows that $\beta \in \mathbb{Z}_p$, $\beta \equiv n_1 \pmod{p^{64}}$, and $p^{64} \mid \mathcal{F}(n_2, n_3, n_4)$.

Since $R(t) = 64 \prod_{b \in B_5}(t - b)$ for $B_5 = \{0, 1, 2401, -343/32, 1/2\}$, every odd old finite contact has odd detector valuation. All differences between these five points are units outside the existing exceptional set. At a

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pole, $v_p(R(t)) = 5v_p(t)$, so an odd infinity depth is excluded by parity soundness. Thus every prime of each $M(t_i)$ supplies the preceding congruences.

If $M(t_i)^{64} \mid N \neq 0$ for all i, then

$$h(a_i) \le 2H|N|^{3/8} + \log 2, \qquad h(x(Q_i)) \le 4H|N|^{3/8} + \tfrac{7}{2}\log 2 < 7H|N|^{3/8}.$$

This is stronger than the bound used in the previous proof. For completeness of the deduction, put $B = \max(|n_1|, |n_2|, |n_3|, |n_4|)$ and $\Delta = \mathcal{F}(n_2, n_3, n_4)$. The inherited scalar and anchor calculations give

$$h(\beta) \le H\sqrt{|n_2|}, \qquad |n_2| > (20H)^2.$$

If $\Delta \neq 0$, then $|\Delta| < 5B^4$ and

$$B^2 - 6 < 7H|\Delta|^{3/8} < 14HB^{3/2},$$

contradicting the anchor. Thus $\Delta = 0$, and the earlier Lucas lemma gives $|n_3| > \exp(|n_2| - 3)$. If $\beta \neq n_1$, apply the same height estimate to $N = |\operatorname{num}(\beta - n_1)| \le 2e^{h(\beta)}B$ to obtain

$$\log B < \tfrac{8}{13}\log(16H) + \tfrac{3}{13}h(\beta) < |n_2| - 3,$$

again a contradiction. Therefore $\beta = n_1 \in \mathbb{Z}$, and $X \in \mathbb{Z}$.

2.4 Completeness and count

For integral X, choose the scalar-height witnesses, anchored S, and Lucas companions exactly as before. Then the eight a_i are fixed rationals. Choose a positive integer D_0 clearing all denominators in $D_0^2 b, D_0^2(a_i - b)$, $b \in B_5$. Apply [1, Theorem 4.1], with $M = 51$, to the distinct positions in the forty-five forms

$$u - bD_0^2w^2, \qquad u - (a_i - b)D_0^2w^2 \quad (1 \le i \le 8,\ b \in B_5).$$

Their coefficients are integers. Choose its lower bounds sufficiently large and set $\eta = u/(D_0w)^2$. The resulting primes are 1 modulo 16 and 51. No selected prime divides $(D_0w)^2$, since otherwise the five distinct prime contacts in its slope would coincide. Each detector is therefore 64 times a product of five distinct positive contact primes divided by $(D_0w)^{10}$.

Orient the places according to Sun's orders d_+, d_- and put $U = \operatorname{lcm} d_+$, $V = \operatorname{lcm} d_-$. The curve in Sun's normalization is isomorphic to ours by scaling with the global unit $3 - 2\sqrt{2}$; its reduction orders are unchanged. Theorem 4.1 gives $\gcd(U, V) = 4$. Choose $k > 4$ with $k \equiv 0 \pmod U$ and $k \equiv 4 \pmod V$, and put $X_0 = kP$, $X_i = kQ_i$. Both exact logarithmic ratios are n_i; hence all tests (5) hold to their prescribed precision at every selected prime. The trace is a unit at each relevant prime, and its parity argument is positive, has square reduced denominator, and is square at $2, 3, 17$. The full preparation predicate supplies all nine main blocks. No uniform real bound or additional coefficient choice is needed.

There are ten chart points, ten four-witness parity blocks, one slope, one correction selector, three anchor witnesses, and one variable for all nonzero guards. Thus the count is

$$10 \cdot 4 + 10 \cdot 4 + 1 + 1 + 3 + 1 = 86.$$

Clear denominators and take the sum of the squared equality numerators, including the single combined inverse equation. This gives one integer polynomial with 86 existential variables besides X. Keeping the two square-root witnesses in each block and the former root-approximation variable instead of trace gives 97 witnesses. The supplied compiler measures degrees and description lengths separately.

2.5 Sharing the scalar-height auxiliary: eighty-two witnesses

One can further substitute $J_0 = Y_0 = X_0 - [4]P$ throughout the scalar-height block and delete its four chart variables. The logarithm denominator already ensures $Y_0 \neq O$. Soundness is a restriction of the preceding formula; a nonzero point of $2E(F)$ admits the deleted chart representation.

For completeness, choose the scalar correction μ first, and let $\mathcal{B}$ be the finite set consisting of $2, 3, 17$ and the prime divisors of $\mu R(\beta)$. Put

$$A_0 = 2 \prod_{p \in \mathcal{B} \setminus \{2\}} p.$$

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Choose a positive multiple T_0 of 64 such that T_0P is sufficiently deep in the formal groups at every place above $\mathcal{B}$. More explicitly, normalize $v_{\mathfrak{p}}(p) = 1$ and arrange $v_{\mathfrak{p}}(f(T_0P)) \geq \max(1, v_p(R(\beta))) + 2$ at every such place. Choose the anchored $S = T_0\Lambda P$ before the ordinary prime pattern, with $v_p(\Lambda) \geq 48\max(1, v_p(R(\beta))) + 2$ for $p \in \mathcal{B}$. The extra two units of precision cover rational coefficient extraction at the ramified prime 2. These choices ensure all scalar-height product factors are units at the necessary primes whenever

$$J_0 = jP, \qquad v_p(j) = v_p(T_0) \quad (p \in \mathcal{B}).$$

Indeed multiplication of T_0P by a p-adic unit preserves its formal depth, and the parameter ratio has the valuation of s/j. The same sufficiently deep neighborhoods apply at bad and ramified places; the required coefficient valuations follow by choosing the initial depths large enough. This is the original scalar-height completeness argument with the auxiliary index valuations fixed in advance.

Now apply Sun's theorem with

$$M = \mathrm{rad}_{\mathrm{odd}}(51T_0A_0)/7^{v_7(\mathrm{rad}_{\mathrm{odd}}(51T_0A_0))}.$$

The two reduction orders are coprime to every odd prime dividing T_0A_0: those other than 7 are excluded by the prescribed M, and the fixed ray condition at 7 gives $d_+ \equiv 6$, $d_- \equiv 2 \pmod 7$. Also $v_2(U) = 2$ and $v_2(V) = 6$. Consequently the three congruences

$$k \equiv 0 \pmod U, \qquad k \equiv 4 \pmod V, \qquad k \equiv 4 + T_0 \pmod{T_0A_0}$$

are compatible: their pairwise common divisors divide 4, 4, and 64, respectively, and $64 \mid T_0$. Choose $k > 4$. Then

$$k - 4 = T_0(1 + A_0b), \qquad v_p(k-4) = v_p(T_0) \quad (p \in \mathcal{B}).$$

Thus the single group constructed from $X_i = kQ_i$ satisfies both the ordinary comparison conditions and the scalar-height conditions with $J_0 = Y_0$. All later parity realizations proceed unchanged. The resulting count is $86 - 4 = 82$ existential variables. The same substitution in the version retaining separate square roots, the root selector, and the three rational Lucas proxies gives 100 variables; it includes the guard $z(Y_0) \neq 0$.

3 A definition with forty existential variables

The following construction uses the height and pole-predicate inputs of Sun's preprint, together with its simultaneous prime-pattern theorem [1, Theorem 4.1]. The latter is an additional arithmetic input; it is not a consequence of the statement that $\mathrm{HTP}_{86}(\mathbb{Q})$ is undecidable. The construction defines the actual subset $\mathbb{Z} \subseteq \mathbb{Q}$.

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3.1 The arithmetic inputs and fixed constants

Put $F = \mathbb{Q}(\sqrt{2})$, let σ be its nontrivial automorphism, and take

$$E : y^2 = x^3 - d^2x, \qquad d = 75 - 53\sqrt{2}.$$

Fix an even positive integer m_0 and a nontorsion point P_0 such that $m_0E(F) = \mathbb{Z}P_0$. We use the following data.

(i) Five distinct rational numbers $\theta_0 = 0, \theta_1, \ldots, \theta_4$, an integer $c \geq 1$, and a fixed exceptional set of rational primes. For every fixed enlargement S of that set, there is $H \geq 1$ such that

$$h(t) \leq HM_S(t)^c \qquad (t \in \mathbb{Q} \setminus \{\theta_0, \ldots, \theta_4\}). \tag{7}$$

Here $M_S(t)$ is the product of primes outside S at which one of the five finite contacts has positive odd normalized depth. More precisely, for primitive integral coordinates $t = (u : v)$, use primitive integral forms proportional to $u - \theta_j v$ and subtract $\min\{v_p(u), v_p(v)\}$. There is *no contact at infinity.* This is the coordinate change and height bound of [1, Section 2.3].

(ii) A finite multiplier list $\mathcal{M} \subset \mathbb{Q}^\times$, a fixed set S_0 containing $2, 3$ and the supports of its elements, a number $\epsilon > 0$, and integers $b_p \geq 1$ for $p \in S_0$ with the following property. For $B \neq 0$, define $\Phi_0(B)$ by a finite choice $\mu \in \mathcal{M}$, two rational structural witnesses e, z, and

$$B' = \mu B, \qquad J_B = B'(B' - B^2), \tag{8}$$

$$BeJ_B z \neq 0, \qquad B'(B' - e)(B' + 3e) \in \mathrm{Sq}(\mathbb{Q}), \tag{9}$$

$$B'(B'z^2 - eJ_B)(B'z^2 + 3eJ_B) \in \mathrm{Sq}(\mathbb{Q}). \tag{10}$$

Then $\Phi_0(B)$ excludes negative odd valuations outside S_0. Conversely it holds whenever B has no negative odd valuation at any prime, $0 < B < \epsilon$, and $v_p(B) \geq b_p$ for every $p \in S_0$. This is [1, Proposition 2.3]. It uses the full square-class realization input of the H10 manuscript; the weaker prime-squareclass input isolated in our third version is not sufficient for this application.

(iii) Given distinct integers $a_1, \ldots, a_k$, an odd squarefree integer M with $7 \nmid M$, a denominator dilation, and arbitrary lower bounds, there are positive integers u, w for which every $u + a_j w^2$ is prime, the quotient u/w^2 is arbitrarily large, and the prescribed dilation divides w. The selected primes split in F and have two reduction orders $d_+(q), d_-(q)$ satisfying

$$v_2(d_+(q)) = 2, \quad v_2(d_-(q)) = 6, \quad \gcd(d_+(q), d_-(q')) = 4 \tag{11}$$

for every pair of selected primes, including an equal pair. Both orders are coprime to M and to 7. These are consequences explicitly included in [1, Theorem 4.1]; the last assertion at 7 follows from the prescribed Gaussian residue $1 + 3i \pmod 7$, giving orders $6, 2 \pmod 7$.

We also use the usual formal group and canonical height of E. Write

$$z(Q) = -x(Q)/y(Q), \quad \rho(Q) = 1/x(Q) \quad (Q \neq O), \qquad z(O) = \rho(O) = 0.$$

These functions are defined on every cyclic point: a nonzero cyclic point is not torsion, so $x(Q)y(Q) \neq 0$. Write ℓ_K for the formal logarithm truncated at degree K. At a good unramified prime $p > K + 1$, two cyclic points aP, bP in the formal kernel, with $b \neq 0$, satisfy

$$v\left(\frac{z(aP)}{z(bP)}\right) = v_p(a/b), \qquad \frac{\ell_K(z(aP))}{\ell_K(z(bP))} \equiv a/b \pmod{p^K} \quad \text{if } v_p(a/b) \geq 0. \tag{12}$$

The valuation of the truncated ratio is always $v_p(a/b)$. At arbitrary fixed places the first identity holds on a sufficiently deep formal subgroup. These are [1, Lemma 2.1 and Section 8.1].

Choose $c_* > 0$, divisible by 64, so that $c_* P_0$ belongs to a deep formal subgroup at all places over S_0, with $v(z(c_* P_0)) \geq 2$. Set $K = 2c$, choose $\xi \in \mathbb{Q} \setminus \{\theta_j\}$, and fix an integer $d_1 > 0$ such that

$$r(T) = d_1^2 \prod_{j=0}^{4} (T - \theta_j) \in \mathbb{Z}[T].$$

Fix an integer $b_* > \max\{1, \theta_0, \ldots, \theta_4\}$. Choose S to contain S_0, all bad and ramified primes, the primes at most $K + 1$, and the supports of c_*, d_1, all contact coefficients and differences, and all $\theta_j - \xi$. Fix H in (7) for this S.

Let $C_E \geq 0$ satisfy $|h(x(Q)) - 2\widehat{h}(Q)| \leq C_E$ on $E(\overline{\mathbb{Q}}) \setminus \{O\}$. Fix an integer $L \geq \lceil H \rceil + 9$ and put

$$C_L = 2L^2 + 6L + 6, \qquad g_L(S, T, W) = T^2 - LS(LS - 2)TW + (LS - 2)^2 W^2 - 1.$$

Replace P_0 by $P = 2^a P_0$ for a fixed nonnegative integer a, and put $m = 2^a m_0$, so that $G = mE(F) = \mathbb{Z}P$ and

$$C := \frac{2H + C_E + \frac{3}{2}\log 2}{2\widehat{h}(P)} < \frac{1}{2\sqrt{C_L}}. \tag{13}$$

No exceptional primes need to be added. In particular $c_* P$ remains in the chosen formal subgroups.

We require one more pair of fixed positive integers A, T_0. Choose A supported on S_0 so that

$$2v_p(A) \geq b_p + 2v_p(2) + 2 \qquad (p \in S_0). \tag{14}$$

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To choose T_0, let $\mathcal{G}$ be the closure of $\mathbb{Z}(P, P^\sigma)$ in $E(\mathbb{R}) \times E^\sigma(\mathbb{R})$. Its identity component is a torus and projects onto the identity circle in each factor. Indeed each projection of $\mathcal{G}$ is an infinite compact subgroup of a real elliptic curve, and $\mathcal{G}$ has finitely many components. In every component choose a nonempty relatively open set on which, at both embeddings,

$$|\rho(Q)|,\ |z(Q)|,\ |1/z(Q)| \le C_0 \tag{15}$$

for one fixed $C_0 \ge 1$. Such sets exist because the forbidden poles and zeros lie over finite subsets of either elliptic factor. Put $C_1 = C_0^2$, and choose T_0 divisible by every prime of S_0 with

$$T_0^4 > A^2(1 + C_1^2)^6/\epsilon. \tag{16}$$

All quantities fixed so far are coefficients or fixed rational functions, not quantified variables. Numerical values have not been emitted here.

3.2 The aggregation and the finite formula

For $R \in \mathbb{Q}^\times$, field expressions $D_j \in F$, and positive integers k_j, let $\mathcal{A}(R; (D_j, k_j)_j)$ denote

$$W = \prod_j \bigl(1 + R(D_j/R^{k_j})^2\bigr), \qquad \Theta = \mathrm{Tr}_{F/\mathbb{Q}}(W^{-1}), \qquad W\Theta \ne 0, \qquad \Phi_0(R/\Theta^2). \tag{17}$$

The displayed quantities are computed expressions. Each occurrence has only the two structural witnesses of Φ_0 and its two finite square-root choices. If $p \notin S$ and $v_p(R)$ is odd, this condition implies that $v_p(R) > 0$ and, at some place $v \mid p$,

$$v(D_j) \ge k_j \quad \text{for every } j. \tag{18}$$

For completeness we will verify the relevant valuations directly. For soundness, put $\alpha = v_p(R)$ and $\lambda_v = v(W) = \sum_j \min\{0, \alpha + 2(v(D_j) - k_j\alpha)\} \le 0$. If $\lambda = \max_{v|p} \lambda_v$, then $v_p(\Theta) \ge -\lambda$. The odd valuation of R/Θ^2 is nonnegative, so $\alpha > 0$ and $\lambda \ge -\alpha/2$. At a place realizing λ, each summand is at least $-\alpha/2$. Hence $v(D_j) \ge (k_j - 3/4)\alpha > k_j - 1$; integral valuations give (18). Zero expressions cause no exception.

For an input $X \in \mathbb{Q}$, put

$$\beta = X^2 + b_*, \qquad r = r(\beta), \qquad N = 1 + T_0^2 r^2, \qquad R_0 = A^2 r.$$

In particular r, N, R_0 are positive. Quantify four nonzero points $Q_n, Q_s, Q_t, Q_w \in G$, a point $H_0 \in G \setminus \{O, c_*P\}$, four points $U_n, U_s, U_t, U_w \in G$, and one rational slope $\eta \ne 0$. The same point H_0 is used in both parts below.

The scalar condition. Set

$$\begin{aligned} W_\beta &= \left(1 + \frac{\rho(Q_s)^2}{R_0}\right)\left(1 + \frac{\rho(H_0)^2}{R_0}\right)\left(1 + \frac{(z(Q_s)/z(H_0))^2}{R_0^{2c-1}}\right), \\ \Theta_\beta &= \mathrm{Tr}_{F/\mathbb{Q}}(W_\beta^{-1}), \qquad W_\beta\Theta_\beta \ne 0, \qquad \Phi_0\left(\frac{R_0}{(N\Theta_\beta)^2}\right). \end{aligned} \tag{19}$$

This is the aggregation (17) for $R = R_0/N^2$ and the three expressions D_j/N^{2k_j-1}, where $(D_j, k_j) = (\rho(Q_s), 1), (\rho(H_0), 1), (z(Q_s)/z(H_0), c)$. The normalizing powers cancel from the product, leaving the shorter displayed formula.

The main conditions. For $i \in \{n, s, t, w\}$ set

$$V_0 = H_0 - c_*P, \quad V_i = U_i - [c_*]Q_i, \qquad L_i = \frac{\ell_K(z(U_i))}{\ell_K(z(H_0))}, \quad L_i' = \frac{\ell_K^\sigma(z(V_i^\sigma))}{\ell_K^\sigma(z(V_0^\sigma))}.$$

Both logarithm denominators are required to be nonzero. Form the following sixteen expressions with their precisions:

expressions	number	precision
$\rho(H_0),\ \rho(U_i)$	5	1
$\rho(V_0^\sigma),\ \rho(V_i^\sigma)$	5	1
$L_i - L_i'$	4	K
$g_L(L_s, L_t, L_w),\ \beta - L_n$	2	K.

(20)

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Denote this list by $(D_j, k_j)_{j=1}^{16}$. Write the eight rational components of the four $x(Q_i)$ as $X_1, \dots, X_8$. Set $a_0 = 1$, and $a_r = X_r^2$ when $X_r \neq 0$, $a_r = 1$ otherwise. For $0 \le r \le 8$ put

$$t_r = a_r \eta, \qquad R_r = a_r^{-5} \frac{\prod_{j=0}^{4}(t_r - \theta_j)}{(t_r - \xi)^6}.$$

Require $t_r \notin \{\theta_0, \dots, \theta_4, \xi\}$ and

$$\mathcal{A}\left(R_r; \left(\frac{D_j}{(t_r - \xi)^{k_j+1}}, k_j\right)_{j=1}^{16}\right) \qquad (0 \le r \le 8). \tag{21}$$

All equalities over F, rational-function evaluations, finite branches and nonzero guards are interpreted on their usual finite rational charts. Equations (10), (17), (19), and (21), together with the point and denominator conditions, are a finite existential formula over $\mathbb{Q}$. No point index is quantified as a rational number.

3.3 Soundness

Lemma 1. *If (19) holds and $Q_s = sP$, then $h(\beta) \le H|s|$.*

Proof. Let $q \mid M_S(\beta)$. At a finite normalized contact of positive depth, β is q-integral and exactly one $\beta - \theta_j$ has positive odd valuation. Thus $v_q(r) > 0$ is odd, N is a q-unit, and $v_q(R_0/N^2) = v_q(r)$ is odd. By (18), at some place over q both points are formal and $v(z(Q_s)/z(H_0)) \ge c$. Writing $H_0 = hP$, (12) gives $v_q(s/h) \ge c$ and hence $q^c \mid s$. Therefore $M_S(\beta)^c \mid s$, and (7) proves the result. If $v_q(\beta) < 0$, all five normalized finite-contact depths are zero: the numerator of each primitive form is a unit. Such a prime does not occur in $M_S(\beta)$. Thus no inference involving the normalization N at a negative valuation is used. □

Lemma 2. *Write $Q_i = n_i P$ and $B = \max_i |n_i|$. If the main conditions hold, then*

$$B^2 \le C|D|^{1/2}$$

whenever either $D = g_L(n_s, n_t, n_w) \neq 0$, or $D = \mathrm{num}(\beta - n_n) \neq 0$.

Proof. Fix $q \mid M_S(t_r)$. Then t_r is integral, exactly one finite contact has positive odd depth, and $t_r - \xi$ is a unit. Since a_r^{-5} is a rational square, $v_q(R_r)$ is odd. Aggregation supplies one place at which the original sixteen expressions in (20) have their specified precisions.

Write $H_0 = hP$, $U_i = z_i P$. The first ten tests make the relevant points formal, and the next four give $L_i - L_i' \in q^K \mathcal{O}_v$. The integers $h, h - c_*$ are nonzero and at least one is a q-unit. Consequently at least one of $z_i/h, (z_i - c_* n_i)/(h - c_*)$ is integral. The valuation identity for truncated logarithms and $L_i - L_i'$ then make both ratios integral. The congruence in (12) applies to both. Multiplying by $h, h - c_*$ and subtracting yields

$$c_* n_i \equiv h L_i - (h - c_*) L_i' \equiv c_* L_i \pmod{q^K \mathcal{O}_v}.$$

Thus $L_i \equiv n_i \pmod{q^K \mathcal{O}_v}$. The final two tests show that q^K divides either nonzero integer D under consideration. For the second test, $\beta - L_n$ first makes β integral at q, so the reduced denominator of $\beta - n_n$ is a q-unit.

Hence $M_S(t_r)^K \mid D$ for every r, and $K = 2c$ gives $h(t_r) \le H|D|^{1/2}$. Since $2h(X_r) = h(a_r) \le h(\eta) + h(a_r \eta)$ for nonzero X_r, every component satisfies $h(X_r) \le H|D|^{1/2}$; a zero component has height zero. For a point whose index has absolute value B,

$$2B^2 \widehat{h}(P) \le h(x(Q_i)) + C_E \le 2H|D|^{1/2} + \tfrac{3}{2}\log 2 + C_E.$$

Use $|D| \ge 1$ and (13). □

Proposition 3. *The displayed formula implies $X \in \mathbb{Z}$.*

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Proof. Write the four indices as n, s, t, w, and put $B = \max\{|n|, |s|, |t|, |w|\}$. They are all nonzero. The integer residual satisfies $|g_L(s,t,w)| \le C_L B^4$. If nonzero, Lemma 2 and (13) give $B^2 < B^2/2$. Hence $g_L(s,t,w) = 0$.

Jones's Lucas classification [3, Corollary 2.7] gives

$$|t| \ge (|Ls| - 1)^{|Ls|-4}.$$

For completeness, write $a = |Ls| \ge 9$ and $y = \operatorname{sgn}(Ls)(Ls-2)w$. The equation becomes $(2t-ay)^2-(a^2-4)y^2 = 4$. Jones's classification and the recurrence $U_0 = 0, U_1 = 1, U_{j+1} = aU_j - U_{j-1}$ give $|y| = U_k$, $|t| \in \{U_{k-1}, U_{k+1}\}$. The congruences $U_k \equiv k \pmod{a-2}$ and $U_k \equiv (-1)^{k-1}k \pmod{a+2}$ imply $k \ge a-2$; $U_j \ge (a-1)^{j-1}$ gives the assertion. Since $L > H$ and $L \ge 9$,

$$\log B \ge (L|s| - 4)\log(L|s| - 1) > H|s|. \tag{22}$$

If $\beta \ne n$, let $D = \operatorname{num}(\beta - n)$. Then $|D| \le 2B \exp h(\beta)$. Lemmas 1 and 2 imply

$$3\log B \le 2\log C + \log 2 + h(\beta) < H|s|,$$

contrary to (22). Thus $\beta = n \in \mathbb{Z}$, so $X^2 = \beta - b_* \in \mathbb{Z}$ and $X \in \mathbb{Z}$. □

3.4 Completeness and sharing the local anchor

We give the order of choices explicitly. Fix $X \in \mathbb{Z}$. Then $\beta \in \mathbb{Z}$, and $r = r(\beta)$ is a positive integer. Let $\mathcal{P}$ be the set of primes dividing Ar. For $p \in \mathcal{P}$ put $\alpha_p = 2v_p(A) + v_p(r) > 0$. The integer $N = 1 + T_0^2 r^2$ is a unit at every such prime: this is immediate at divisors of r, and at divisors of A it follows from the divisibility imposed on T_0.

Choose a positive integer λ such that

$$v_p(\lambda) \ge c\alpha_p + v_p(2) + 1 \qquad (p \in \mathcal{P}).$$

Choose a positive integer D_0, divisible by c_*, so that every point in $D_0 G$ belongs, at all places over $\mathcal{P}$, to a sufficiently deep formal subgroup and satisfies $v(\rho(Q)) \ge \alpha_p + v_p(2) + 1$ unless $Q = O$. Such a D_0 exists by the finite index of deep local formal subgroups. Choose a positive multiple j_0 of D_0 for which $Q_s = [\lambda j_0]P$ belongs to one of the open sets (15) at the real places. This is possible with arbitrarily large j_0: the closure of any nonconstant arithmetic progression in $\mathcal{G}$ contains an entire component, because multiplication by a nonzero integer is surjective on its identity torus. The positive tail of the progression has the same closure.

Put $s = \lambda j_0$, $n = \beta$, and choose positive t, w satisfying $g_L(s,t,w) = 0$. Explicitly take the Lucas sequence with parameter Ls, put $k = Ls - 2$, and take $t = U_{k+1}$ and $w = U_k/k$. The congruence $U_k \equiv k \pmod{Ls-2}$ and the identity $U_{k+1}^2 - LsU_{k+1}U_k + U_k^2 = 1$ verify these choices. All four root points are now fixed.

Set

$$D_1 = \operatorname{lcm}\left(D_0, \prod_{p \in \mathcal{P}} p^{v_p(j_0)+1}\right). \tag{23}$$

We will require the shared index h to satisfy $h \equiv j_0 \pmod{D_1}$. Then $H_0 = hP$ has the same formal depths as $j_0 P$ at the places in $\mathcal{P}$, and

$$v(\rho(H_0)), v(\rho(Q_s)) \ge \alpha_p + v_p(2) + 1, \quad v(z(Q_s)/z(H_0)) = v_p(\lambda) \ge c\alpha_p + v_p(2) + 1. \tag{24}$$

Indeed both $hP, j_0 P$ lie in the chosen deep formal subgroup and $v_p(h) = v_p(j_0)$, so the formal logarithm compares their parameter valuations. Also $c_* \mid h$.

We additionally choose H_0 in an open set from (15). Then the three unnormalized scalar expressions have absolute values at most C_1 at both embeddings. At these embeddings the three nonconstant terms in the scalar product (17) have the exact form

$$D_j^2 A^{2-4k_j} r^{1-2k_j}, \qquad (D_j, k_j) = (\rho(Q_s), 1), (\rho(H_0), 1), (z(Q_s)/z(H_0), c).$$

They are nonnegative and at most C_1^2. Consequently $1 \le W_\beta \le (1 + C_1^2)^3$, $\Theta_\beta \ge 2/(1 + C_1^2)^3$, and

$$0 < \frac{R_0}{(N\Theta_\beta)^2} \le \frac{A^2(1 + C_1^2)^6}{4T_0^4} < \epsilon.$$

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At $p \in \mathcal{P}$, (24) gives valuation at least $\alpha_p + 2v_p(2) + 2$ for each nonconstant term. Thus $\Theta_\beta = 2 +$ an element of valuation $> v_p(2)$ and

$$v_p\left(R_0/(N\Theta_\beta)^2\right) = \alpha_p - 2v_p(2).$$

This is at least b_p on S_0 by (14), and is nonnegative at every other prime of $\mathcal{P}$. Outside $\mathcal{P}$ the valuation of R_0/N^2 is even, so that of $R_0/(N\Theta_\beta)^2$ is even. The input therefore has no negative odd valuation at any prime. The completeness assertion for Φ_0 proves the scalar condition, for every later h having the stipulated congruence and real bounds.

It remains to construct such an h together with the main slope. Set $U_i = [n_i]H_0$ for $i = n, s, t, w$ in the completeness construction. Then $V_i = [n_i]V_0$. The basic reciprocal coordinates and each individual truncated-logarithm ratio are rational functions on one real elliptic factor. Their poles and logarithm-denominator zeros lie over finite subsets of that factor. Bounds for these basic functions bound all sixteen expressions in (20). None of the forbidden finite fibres contains a component of $\mathcal{G}$. Inside the previously chosen open set in each component, remove these fibres and choose a smaller nonempty open set. On these smaller sets all sixteen main expressions have a common bound C_2. Crucially C_2 is chosen before the main prime pattern and is independent of its later congruence moduli.

At every place over S_0, the sixteen main expressions are integral whenever $c_* \mid h$ and $h \notin \{0, c_*\}$. Indeed $hP, (h - c_*)P$ lie in the fixed deep formal subgroup. Their integer multiples give truncated logarithm ratios of valuation $v_p(n_i) \geq 0$, and their reciprocal x-coordinates are integral. Since β and the coefficients of g_L are integers, the last two expressions are integral as well.

For the now fixed positive squares a_r, choose a square denominator $v = d_2^2 w^2$ with d_2 sufficiently divisible and put $\eta = u/v$. Clear denominators so that all distinct positions

$$u - (\theta_j/a_r)v \quad (0 \leq r \leq 8,\ 0 \leq j \leq 4) \tag{25}$$

are integral quadratic polynomials in u, w. Since $\theta_0 = 0$, there are at most 37 distinct positions. Apply the prime-pattern input with the odd squarefree product of the primes dividing D_1, except 7, as its avoidance modulus. Take all selected primes beyond the fixed exceptions and the supports and differences of these rational coefficients. The theorem also permits arbitrarily large η and arbitrary fixed denominator divisibility.

Let U be the least common multiple of all selected $d_+(q)$ and V that of all selected $d_-(q)$. Impose

$$h \equiv 0 \pmod{U}, \qquad h \equiv c_* \pmod{V}, \qquad h \equiv j_0 \pmod{D_1}. \tag{26}$$

These congruences are compatible. Indeed $\gcd(U, V) = 4 \mid c_*$. Every common divisor of UV and D_1 is a power of two: the avoidance condition handles all odd primes other than 7, and the fixed Gaussian residue handles 7. The relevant powers of two divide 4 for U and 64 for V. Both j_0 and c_* are divisible by 64. This checks every pair of congruences, including arbitrary powers in D_1. The generalized Chinese remainder theorem therefore gives a nonconstant arithmetic progression of solutions. Choose an arbitrarily large positive solution whose real point belongs to one of the smaller open sets defining C_2; avoid $h = c_*$.

At each selected contact prime orient the two places so that the first has order $d_+(q)$. Then H_0, U_i are formal at the first place and V_0, V_i at the second. The two rational index ratios in the formal comparison are both exactly n_i. Thus all sixteen main expressions have the required precisions at the first place: conjugation supplies the second family there, the logarithm differences vanish modulo q^K, and $g_L(n_s, n_t, n_w) = \beta - n_n = 0$.

For completeness of the pole predicates, the exact detector formula is

$$R_r = \frac{\prod_{j=0}^{4}\left(u - (\theta_j/a_r)v\right)}{v^5(t_r - \xi)^6}.$$

Within each slope its five contact primes are distinct; v is a square and is a unit at those primes. Hence $v_q(R_r) = 1$ at its five contacts, and $v_p(R_r)$ is even elsewhere. At a contact, every local factor in W is either 1 modulo its place or has negative valuation. At the successful place they are all 1 modulo that place. Consequently the trace of W^{-1} has residue 1 or 2, which is nonzero, so the pole input has valuation one there and even valuation elsewhere. It has no negative odd valuation.

Here are the bounds ensuring the remaining preparation requirements. Uniformly when the sixteen unscaled expressions are bounded by C_2, as $\eta \to +\infty$,

$$R_r = a_r^{-6}\eta^{-1}(1 + O(\eta^{-1})), \quad W_r = 1 + O(\eta^{-3}), \quad \Theta_r = 2 + O(\eta^{-3}),$$
$$R_r/\Theta_r^2 = (4a_r^6\eta)^{-1}(1 + O(\eta^{-1})).$$

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Thus all nine pole inputs are positive and less than ϵ once a fixed lower bound on η is imposed before the pattern is selected. For $p \in S_0$ put $\nu = v_p(v)$ and $\gamma_r = v_p(a_r)$, both even. For sufficiently divisible v, the prime u is a p-unit and

$$v_p(R_r) = \nu - 6\gamma_r, \qquad v\left(R_r\left(\frac{D_j/(t_r-\xi)^{k_j+1}}{R_r^{k_j}}\right)^2\right) \geq 3\nu + (10k_j - 8)\gamma_r.$$

All unscaled D_j are integral. Increasing the prescribed divisibility of v makes the right side exceed $v_p(2)$ for every r, j. It follows that $v_p(\Theta_r) = v_p(2)$ and $v_p(R_r/\Theta_r^2) = \nu - 6\gamma_r - 2v_p(2) \geq b_p$. All bounds were fixed before the prime pattern and its CRT modulus. The completeness assertion for Φ_0 now verifies every main predicate. This completes the proof that the formula holds for every integer X.

3.5 Polynomial compilation and variable count

Theorem 4. *Assume the arithmetic inputs above. There is a polynomial $F \in \mathbb{Z}[X, Y_1, \ldots, Y_{40}]$ such that, for every $X \in \mathbb{Q}$,*

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists Y_1, \ldots, Y_{40} \in \mathbb{Q} \quad F(X, \mathbf{Y}) = 0.$$

Given numerical realizations of the fixed constants and finite multiplier data above, the proof gives a terminating finite algorithm to produce F. It does not claim that this polynomial, its circuit, or its coefficients have been emitted here.

Proof. The equivalence was proved above. There are nine cyclic points: four root points and five shared local points. Represent each as $[m](x, y)$ with $(x, y) \in E(F)$ affine. Every cyclic point admits such an affine preimage; for O, use the two-torsion point $(0, 0)$. Writing $x = u + v\sqrt{2}$ gives two free rational coordinates. If $x^3 - d^2x = C_0 + C_1\sqrt{2}$ and $C_1 \neq 0$, its ordinate is encoded by the single finite choice

$$\tau^4 - C_0\tau^2 + C_1^2/2 = 0, \qquad y = \tau + \frac{C_1}{2\tau}\sqrt{2}.$$

This quartic is separable and $\tau \neq 0$. If $C_1 = 0$, use the finite branches $y = \tau$, $\tau^2 = C_0$, or $y = \tau\sqrt{2}$, $\tau^2 = C_0/2$; split off $C_0 = 0$. Thus each point contributes two free coordinates and one finite algebraic choice.

There are ten pole predicates: nine main predicates and one scalar predicate. Each contributes two structural coordinates e, z and two finite square-root choices. There is one free slope η. Consequently the free structural parameters number

$$9 \cdot 2 + 10 \cdot 2 + 1 = 39.$$

The finite-algebra compilation of [1, Section 10] combines every remaining finite choice, branch and nonzero guard into one final rational variable. For clarity, it first separates the points of each finite reduced algebra by one of finitely many fixed linear combinations, retains a nonzero discriminant, recovers the component roots by matrix inversion, and filters equalities and inequations by polynomial gcds. Dependent square-root choices are then incorporated in a second reduced algebra and separated again. Degree drops and zero square inputs are handled by finite branches. Finally the reciprocal-polynomial construction absorbs every denominator and nonzero guard into that same variable; sums of squares combine equations and products combine the finitely many branches. This is an algorithm on polynomials of fixed finite degree and uses no additional free parameters. It gives $39 + 1 = 40$ existential variables. □

There is a less compressed presentation that avoids the global finite-algebra step. Keep one ordinate variable for each point. For the two square tests in a pole predicate, use the elementary identity

$$A, B \in \mathrm{Sq}(\mathbb{Q}) \quad \Longleftrightarrow \quad \exists t \in \mathbb{Q} : \quad (A^2 + B^2)\big(1 - 2(A+B)t^2 + (A-B)^2t^4\big) = 0.$$

Indeed, when not both inputs vanish, a rational root gives a nonzero rational number equal to a signed sum of square roots, from which both square roots are recovered using their sum and difference; the converse chooses a nonzero signed sum. This leaves

$$9 \cdot 3 + 10 \cdot 3 + 1 + 1 = 59$$

variables, the last enforcing the product of all nonzero guards on each finite branch. The formula consists of the displayed rational-function compositions, local quartics and finite branches. It is likely more useful for a short

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circuit description than the forty-variable polynomial. Before branching, the uncompressed finite fibres already have degree at most $4^9 2^{20} = 2^{38}$, so global primitive-element packing can increase degree and coefficient data enormously. Neither count alone establishes an improvement in total encoded bit length.

4 Scope of the audits

The geometric-sieve audit checks Sun's Appendix A and Section 4 against Browning–Heath-Brown's Lemmas 2.1 and 4.1 and Browning–Loughran's Theorem 4.1. In particular, the latter permits five variables, and the nonhomogeneous shifted norms are treated by the appropriate chart arguments. The analytic audit checks Sections 5–6 against the exact linear Artin-symbol theorem of Frei–Tinková and the dense-model and polynomial counting results of Tao–Ziegler. It also checks the order of limits: the obstruction cutoff and ray field are fixed before the analytic limit, and the upper-bound sieve constant is independent of that cutoff. No concrete gap was found in these arguments.

Independent mathematical checks cover the grouped contact detector, shared-point congruences, the two-logarithm argument, the scalar certificate, the final Lucas contradiction and finite-algebra count. The circuit audit recounts variables, operations and degree without importing the builder, checks integer coefficients and the sum-of-squares root, and verifies a lossless encoding round trip. It does not expand the high-degree polynomial. Detailed audit records and reproducible code accompany this article.

The explicit construction imports the earlier curve, height bound and simultaneous-twist realization assertions in addition to Sun's theorem. The forty-variable construction imports the height and full square-class realization inputs used in Sun's Section 2. It therefore uses stronger realization data than the restricted input isolated in our third article. Accepting those stated inputs makes the results conditional theorems; the audit does not replace those hypotheses by an assertion of independent verification of the entire H10 manuscript.

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